

TrackDancer — Shopping List (two-board build)

Companion to the v2 board sheet. Check the OWNED list at the bench first — buy only what's missing.

0. Verify on hand before shopping

- ☐ Resistor kit has plenty of 2.2k and 3.3k
- ☐ Hookup wire (solid 22-24 AWG + some stranded)
- ☐ Cobbler ribbon cable located
- ☐ Pololu D24V50F5 buck located (bench-test it soon)
- ☐ Zip ties / heat shrink / epoxy or hot glue
- ☐ Spare USB micro cable for the Teensy

1. Board build

item	qty	~\$	notes
☐ FR-4 perfboard, double-sided plated-through ~7x9cm	3-pack	8	A + B + spare
☐ Screw terminal blocks, PCB mount (3.5 or 5.08mm)	20 pos	8	12+ on B, bus x2
☐ Female header socket strips, 2.54mm cuttable	10-pk	8	socket Teensy + modules
☐ 2x20 IDC box header (ribbon landing)	1	2	or solder cores direct
☐ Brass standoff kit M2.5 + M3	kit	10	Pi=M2.5, boards=M3
☐ Heat-set threaded inserts M3 (+ few M2.5)	kit	12	for the PA612-CF15
☐ Pi 5 official active cooler	1	8	non-negotiable in a car box
☐ 100nF ceramic capacitors	10+	3	one per divider output

2. Power and 12V interface

item	qty	~\$	notes
☐ Inline blade fuse holder + 5A fuses	1 + spares	6	12V feed to the Pololu
☐ XT30 pair (or barrel pair) for 12V input	1 pair	4	same plug bench + car
☐ Ring terminal / crimp assortment	kit	6	install-day 12V + grounds

Estimated total new spend: \$55-75

3. NOT today — buy when its build starts

- ☐ AEM 30-2012 oil temp sender (thermistor build) ~45
- ☐ u-blox GPS 10-25Hz (lap timing / track map) ~40
- ☐ Deutsch DT 12-pin bulkhead kit (car harness) ~25
- ☐ UVC camera with ONBOARD H.264 (video phase) varies
- ☐ BAT54S schottky clamps (hardening pass) ~5

Ignition-sense / clean-shutdown parts: decide the circuit first — before real driving.